

Lab6

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```
In [1]: !pip install graphframes
```

Requirement already satisfied: graphframes in /usr/local/lib/python3.10/dist-packages (0.6)

Requirement already satisfied: nose in /usr/local/lib/python3.10/dist-packages (from graphframes) (1.3.7)

Requirement already satisfied: numpy in /usr/local/lib/python3.10/dist-packages (from graphframes) (2.2.6)

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

```
In [ ]: from spark_utils import SparkUtils
neo4j_connector = "org.neo4j:neo4j-connector-apache-spark_2.13:5.3.10_for_spark_3,io.graphframes:graphframes-spark3_2.13:0.9.0"
su = SparkUtils("Lab 6", "spark://spark-master:7077", spark_packages=neo4j_connector)
su._spark
```

```
WARNING: Using incubator modules: jdk.incubator.vector
:: loading settings :: url = jar:file:/opt/spark/jars/ivy-2.5.3.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /root/.ivy2.5.2/cache
The jars for the packages stored in: /root/.ivy2.5.2/jars
org.neo4j#neo4j-connector-apache-spark_2.13 added as a dependency
io.graphframes#graphframes-spark3_2.13 added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-eabe4e2f-b008-43ae-b404-8bee2a6654e8;1.0
  confs: [default]
    found org.neo4j#neo4j-connector-apache-spark_2.13;5.3.10_for_spark_3 in central
    found org.neo4j#neo4j-connector-apache-spark_2.13_common;5.3.10_for_spark_3 in central
    found org.neo4j#caniuse-core;1.3.0 in central
    found org.neo4j#caniuse-api;1.3.0 in central
    found org.jetbrains.kotlin#kotlin-stdlib;2.1.20 in central
    found org.jetbrains#annotations;13.0 in central
    found org.neo4j#caniuse-neo4j-detection;1.3.0 in central
    found org.neo4j.driver#neo4j-java-driver-slim;4.4.21 in central
    found org.reactivestreams#reactive-streams;1.0.4 in central
    found io.netty#netty-handler;4.1.127.Final in central
    found io.netty#netty-common;4.1.127.Final in central
    found io.netty#netty-resolver;4.1.127.Final in central
    found io.netty#netty-buffer;4.1.127.Final in central
    found io.netty#netty-transport;4.1.127.Final in central
    found io.netty#netty-transport-native-unix-common;4.1.127.Final in central
    found io.netty#netty-codec;4.1.127.Final in central
    found io.netty#netty-tcnative-classes;2.0.73.Final in central
    found io.projectreactor#reactor-core;3.6.11 in central
    found org.neo4j#neo4j-cypher-dsl;2022.11.0 in central
    found org.apiguardian#apiguardian-api;1.1.2 in central
    found org.neo4j.connectors#commons-authn-spi;1.0.0-rc2 in central
    found org.neo4j.connectors#commons-reauth-driver;1.0.0-rc2 in central
    found org.slf4j#slf4j-api;2.0.17 in central
    found org.neo4j.connectors#commons-authn-provided;1.0.0-rc2 in central
    found io.graphframes#graphframes-spark3_2.13;0.9.0-spark3.5 in central
:: resolution report :: resolve 1275ms :: artifacts dl 39ms
  :: modules in use:
    io.graphframes#graphframes-spark3_2.13;0.9.0-spark3.5 from central in [default]
    io.netty#netty-buffer;4.1.127.Final from central in [default]
    io.netty#netty-codec;4.1.127.Final from central in [default]
    io.netty#netty-common;4.1.127.Final from central in [default]
    io.netty#netty-handler;4.1.127.Final from central in [default]
```

```

io.netty#netty-resolver;4.1.127.Final from central in [default]
io.netty#netty-tcnative-classes;2.0.73.Final from central in [default]
io.netty#netty-transport;4.1.127.Final from central in [default]
io.netty#netty-transport-native-unix-common;4.1.127.Final from central in [default]
io.projectreactor#reactor-core;3.6.11 from central in [default]
org.apiguardian#apiguardian-api;1.1.2 from central in [default]
org.jetbrains#annotations;13.0 from central in [default]
org.jetbrains.kotlin#kotlin-stdlib;2.1.20 from central in [default]
org.neo4j#caniuse-api;1.3.0 from central in [default]
org.neo4j#caniuse-core;1.3.0 from central in [default]
org.neo4j#caniuse-neo4j-detection;1.3.0 from central in [default]
org.neo4j#neo4j-connector-apache-spark_2.13;5.3.10_for_spark_3 from central in [default]
org.neo4j#neo4j-connector-apache-spark_2.13_common;5.3.10_for_spark_3 from central in [default]
org.neo4j#neo4j-cypher-dsl;2022.11.0 from central in [default]
org.neo4j.connectors#commons-authn-provided;1.0.0-rc2 from central in [default]
org.neo4j.connectors#commons-authn-spi;1.0.0-rc2 from central in [default]
org.neo4j.connectors#commons-reauth-driver;1.0.0-rc2 from central in [default]
org.neo4j.driver#neo4j-java-driver-slim;4.4.21 from central in [default]
org.reactivestreams#reactive-streams;1.0.4 from central in [default]
org.slf4j#slf4j-api;2.0.17 from central in [default]

```

```

-----
|               | modules                | artifacts |
|      conf     | number| search|dwnlded|evicted|| number|dwnlded|
|-----|-----|-----|-----|-----|
|      default  | 25   | 0    | 0    | 0    || 25   | 0    |
|-----|-----|-----|-----|-----|

```

```
:: retrieving :: org.apache.spark#spark-submit-parent-eabe4e2f-b008-43ae-b404-8bee2a6654e8
```

```
  confs: [default]
```

```
  0 artifacts copied, 25 already retrieved (0kB/24ms)
```

```
26/03/23 15:34:26 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

```
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
```

```
Setting default log level to "WARN".
```

```
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
```

Out[]: **SparkSession - in-memory**

SparkContext

[Spark UI](#)

Version v4.0.1

Master spark://spark-master:7077

AppName Lab 6

Data Ingestion

```
In [3]: videogames_schema = SparkUtils.generate_schema([("user", "string"), ("videogame", "string"), ("rating", "float"), ("timestamp", "timestamp")])

videogames_df = su._spark \
    .read \
    .option("header", "false") \
    .option("sep", ",") \
    .schema(videogames_schema) \
    .csv("/opt/spark/work-dir/data/videogames/")

videogames_df.show()
```

[Stage 0:>

(0 + 1) / 1]

```
+-----+-----+-----+-----+
|      user| videogame| rating| timestamp|
+-----+-----+-----+-----+
| AB9S92790Z3Q0|0078764343|  5.0|1373155200|
| A24SSUT5CSW8BH|0078764343|  5.0|1377302400|
| AK3V0HEBJMQ7J|0078764343|  4.0|1372896000|
| A10BECPH7W8HM7|043933702X|  5.0|1404950400|
| A2PRV90ULX1TWP|043933702X|  5.0|1386115200|
| AE7GUHCDQ4UI|043933702X|  1.0|1366156800|
| A48ABFDDRMKI8|043933702X|  5.0|1374192000|
| A26B0P6K95SIKW|0439339960|  3.0|1288569600|
| AZ3UWOC8QSO6C|0439339987|  5.0|1366848000|
| A182S3ANC0W7DL|0439342260|  4.0|1355875200|
| AY5Q951JAZAX9|0439374391|  5.0|1356393600|
| A1TL721YECDIM8|0439394422|  3.0|1389657600|
| APDCEJMFDO2YT|0439394422|  5.0|1285545600|
| AFJ7A9CSEPZNY|043940133X|  4.0|1168300800|
| A14YVGE643TRJK|043940133X|  1.0|1324944000|
| A2H3TQWU51W1WE|043940133X|  5.0|1299456000|
| A3K010N2ODLHBR|043940133X|  5.0|1258329600|
| A1T98OCCYW60BI|043940133X|  2.0|1343779200|
| A3P758JGM12DD7|043940133X|  5.0|1357862400|
| A1TBUSGCBTXWFC|043940133X|  1.0|1256428800|
+-----+-----+-----+-----+
```

only showing top 20 rows

Graph Analysis

Build GraphFrame

```
In [4]: from graphframes import GraphFrame
        from pyspark.sql.functions import col

        users = videogames_df.select(col("user").alias("id")).distinct()
        products = videogames_df.select(col("videogame").alias("id")).distinct()
        vertices = users.union(products).distinct()
```

```
edges = videogames_df.select(
    col("user").alias("src"),
    col("videogame").alias("dst"),
    col("rating")
)

g = GraphFrame(vertices, edges)
print(f"Vertices: {g.vertices.count()}")
print(f"Edges: {g.edges.count()}")
```

/opt/spark/python/pyspark/sql/classic/dataframe.py:146: UserWarning: DataFrame.sql_ctx is an internal property, and will be removed in future releases. Use DataFrame.sparkSession instead.

warnings.warn(

Vertices: 876977

[Stage 15:=====> (1 + 1) / 2]

Edges: 1324753

PageRank

In [28]: results = g.pageRank(resetProbability=0.15, maxIter=10)

```
results.vertices \
    .select("id", "pagerank") \
    .orderBy("pagerank", ascending=False) \
    .show()
```

[Stage 1215:=====> (4 + 1) / 5]

```

+-----+-----+
|      id|      pagerank|
+-----+-----+
|B00DJFIMW6| 7285.526431395484|
|B00BGA9WK2| 2645.507195083908|
|B00FAX6XQC| 2547.3341328377564|
|B009KS4XRO| 2501.0385889444806|
|B0055SWM08| 1989.4563258041183|
|B00CSR2J9I| 1964.373283273377|
|B002VBWIP6| 1857.1934441848766|
|B0015AARJI| 1408.3694893308366|
|B000FKBCX4| 1243.5554664378442|
|B00178630A| 1230.274656615952|
|B007VTVRFA| 1206.0855893139276|
|B00AIUUXHC| 1173.889544941537|
|B003VANOFY| 986.3206066999797|
|B007FTE2VW| 984.3404153958518|
|B000B9RI14| 933.3473057681464|
|B00EF10GOG| 813.309107754935|
|B005EQE0YM| 789.8962020979171|
|B0009VXBAQ| 759.9537257156875|
|B0029LJIFG| 752.2411136270933|
|B00AW3FDIO| 740.7225573743946|
+-----+-----+
only showing top 20 rows

```

Label Propagation

```
In [29]: lpa = g.labelPropagation(maxIter=5)
lpa.show()
```

```
[Stage 1444:=====> (4 + 1) / 5]
```

```

+-----+-----+
|          id|      label|
+-----+-----+
|    043933702X|25769804420|
|    0439671418| 8589972610|
|    0439900581|17179871748|
|    0545115507|      55201|
|    0700026657| 8590005570|
|    1886846758|17179911574|
|    7293000936| 8590098049|
|    7542614444|25769958148|
|    9078439122|34359872010|
|    9861064222|17179908478|
|    986118452X| 8590070871|
|    9861767304| 8589942402|
|    986325083X|25769883582|
|    9941113300|      94401|
|A0009878M2RGMMHGJH39|      169495|
|A00101961G0VS92WD...| 8590100259|
|A001147626R4BL248...|25769977765|
|A0011756FPL8K71Q5TAQ|34359907769|
|A00230923E4Y7VHWZ...| 8590107813|
|A00338543M2OZPUWO...|      172460|
+-----+-----+
only showing top 20 rows

```

Triangle Counting

```
In [30]: triangle_count = g.triangleCount()
triangle_count.show()
```

```
[Stage 1581:=====> (4 + 1) / 5]
```



```

+-----+-----+
| count |          id |
+-----+-----+
| 0 | A24SSUT5CSW8BH |
| 0 | AE7GUHCDQQ4UI |
| 0 | AY5Q951JAZAX9 |
| 0 | APDCEJMFDO2YT |
| 0 | A14YVGE643TRJK |
| 0 | A1TBUSGCBTXWFC |
| 0 | A221V1CP1DVGRI |
| 0 | A129SW886TYQ6H |
| 0 | AFWPLXT20D6H1 |
| 0 | A34UGG0WDKRTQF |
| 0 | A1WJW20IR5BZ58 |
| 0 | A1VJVU8YSBPAY7 |
| 0 | A11DR0XTI0P0L |
| 0 | A1JPYSVX02PV36 |
| 0 | A1EO9BFUHTGWKZ |
| 0 | A6Q1IUFSN2E2D |
| 0 | A1LMJ9W8UX1H5B |
| 0 | A3GB90F65GKZF7 |
| 0 | A1125BZR4DT8RR |
| 0 | A1NCF8YOK4YVY8 |
+-----+-----+
only showing top 20 rows

```

Degree Distribution

InDegree

```

In [31]: in_deg = g.inDegrees.join(vertices, "id")
         in_deg.orderBy("inDegree", ascending=False).show()

```

[Stage 1597:=====>

(2 + 1) / 3]

```

+-----+-----+
|      id | inDegree |
+-----+-----+
| B00DJFIMW6 | 16221 |
| B00BGA9WK2 | 7561 |
| B00FAX6XQC | 5713 |
| B009KS4XRO | 5489 |
| B002VBWIP6 | 5190 |
| B0055SWM08 | 4638 |
| B00CSR2J9I | 4510 |
| B0015AARJI | 4468 |
| B00178630A | 3522 |
| B000FKBCX4 | 3290 |
| B007VTVRFA | 3122 |
| B000B9RI14 | 2996 |
| B007FTE2VW | 2743 |
| B00AIUUXHC | 2739 |
| B003VANOFY | 2572 |
| B0009VXBAQ | 2409 |
| B0050SYX8W | 2346 |
| B0029LJIFG | 2309 |
| B00CMQTVK0 | 2219 |
| B005EQE0YM | 2194 |
+-----+-----+
only showing top 20 rows

```

OutDegree

```

In [32]: out_deg = g.outDegrees.join(vertices, "id")
         out_deg.orderBy("outDegree", ascending=False).show()

```

[Stage 1608:=====>

(2 + 1) / 3]

```

+-----+-----+
|          id | outDegree |
+-----+-----+
| A3V6Z4RCDGRC44 |      880 |
| A3W4D8XOGLWUN5 |      817 |
|  AJKWF4W7QD4NS |      797 |
| A2QHS1ZCIQOL7E |      521 |
| A2TCG2HV1VJP6V |      474 |
| A29BQ6B90Y1R5F |      429 |
|  AFV2584U13XP3 |      338 |
| A20DZX38KRBIT8 |      320 |
|  A74TA8X5YQ7NE |      267 |
| A2582KMXLK2P06 |      263 |
| A3GKMQL05Z79K |      253 |
| A3J8ABVGK7ZL6H |      250 |
|  AQMWZIH22R6LE |      239 |
| A1AISPOIIHTHXX |      227 |
| A1LBAC84TLIGAX |      224 |
| A2GBBDNZLYC4A9 |      222 |
|  A8NHN9UPML858 |      217 |
|  AWG209C42XW5G |      212 |
|  ANAYSRE3LX8GZ |      210 |
| A319SKSB556033 |      209 |
+-----+-----+
only showing top 20 rows

```

Writing Data in Neo4j

```

In [5]: neo4j_url = "bolt://neo4j-iteso:7687"
        neo4j_user = "neo4j"
        neo4j_passwd = "neo4j@1234"

```

```

In [7]: neo4j_options = {
        "url": neo4j_url,
        "authentication.basic.username": neo4j_user,
        "authentication.basic.password": neo4j_passwd,
        "batch.size": "10000",

```

```
"transaction.retries": "3"
}
```

In [8]: `from pyspark.sql.functions import col`

```
user_vertices = videogames_df.select(col("user").alias("id")).distinct()

user_vertices.write \
    .format("org.neo4j.spark.DataSource") \
    .mode("Overwrite") \
    .options(**neo4j_options) \
    .option("labels", "User") \
    .option("node.keys", "id") \
    .save()

print(f"{user_vertices.count()} user vertices wrote in Neo4j")
```

[Stage 23:=====> (2 + 1) / 3]
826767 user vertices wrote in Neo4j

In [9]: `product_vertices = videogames_df.select(col("videogame").alias("id")).distinct()`

```
product_vertices.write \
    .format("org.neo4j.spark.DataSource") \
    .mode("Overwrite") \
    .options(**neo4j_options) \
    .option("labels", "Videogame") \
    .option("node.keys", "id") \
    .save()

print(f"{product_vertices.count()} videogame vertices wrote in Neo4j")
```

[Stage 30:=====> (1 + 1) / 2]
50210 videogame vertices wrote in Neo4j

```
In [10]: rating_edges = videogames_df.select(
    col("user").alias("src_id"),
    col("videogame").alias("dst_id"),
    col("rating"),
    col("timestamp")
)
rel_query = """
MATCH (u:User {id: event.src_id})
MATCH (v:Videogame {id: event.dst_id})
MERGE (u)-[r:RATED]->(v)
SET r.rating = event.rating,
    r.timestamp = event.timestamp
"""

rating_edges.write \
    .format("org.neo4j.spark.DataSource") \
    .mode("Append") \
    .options(**neo4j_options) \
    .option("query", rel_query) \
    .save()

print(f"{rating_edges.count()} edges wrote in Neo4j")
su.spark.stop()
```

1324753 edges wrote in Neo4j

```
In [11]: su.spark.stop()
```

```
In [ ]:
```